

GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB

OBJECTIVES:

- (i) To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.
- (ii) (ii) To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

OBJECTIVE 1: To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

PROGRAM:

```
clc;

clear;

close all;

% Parameters

Am = 2;

Ac = 1;

fm = 5;

fc = 50;

beta = 5;

fs = 2000;

% Time vector

t = 0:1/fs:1;
```

```
% Message signal
```

```
m = Am * cos(2*pi*fm*t);
```

```
% Carrier signal
```

```
c = Ac * cos(2*pi*fc*t);
```

```
% FM signal
```

```
fm_sig = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
```

```
% Plot signals
```

```
figure('Name','Frequency Modulation (FM)','NumberTitle','off');
```

```
subplot(3,1,1)
```

```
plot(t,m)
```

```
title('Message Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
subplot(3,1,2)
```

```
plot(t,c)
```

```
title('Carrier Signal')
```

```
xlabel('Time (s)')
```

```
ylabel('Amplitude')
```

```
grid on
```

```
subplot(3,1,3)
```

```
plot(t,fm_sig)
```

```

title('FM Signal')

xlabel('Time (s)')

ylabel('Amplitude')

grid on

% Display parameters

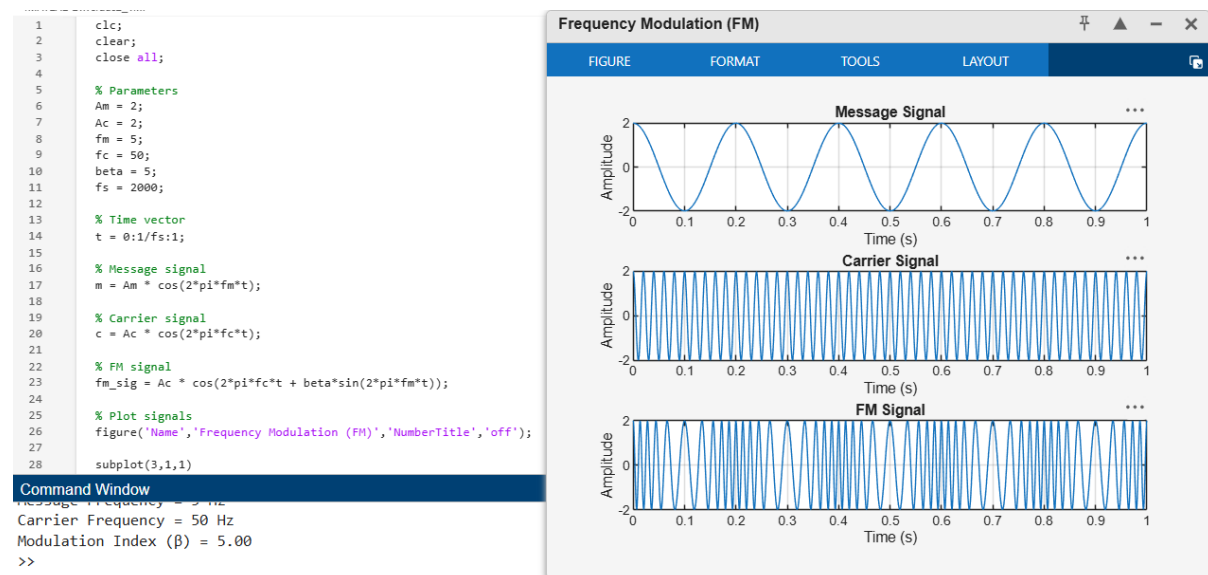
fprintf('Message Frequency = %d Hz\n', fm);

fprintf('Carrier Frequency = %d Hz\n', fc);

fprintf('Modulation Index ( $\beta$ ) = %.2f\n', beta);

```

OUTPUT:



OBJECTIVE 2 : To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

PROGRAM:

```

clc;

clear;

close all;

```

%% Input Parameters

Am = 2; % Message signal amplitude

fm = 500; % Message frequency (Hz)

Ac = 1; % Carrier amplitude

fc = 10000; % Carrier frequency (Hz)

kf = 1000; % Frequency sensitivity (Hz/V)

fs = 100000; % Sampling frequency (Hz)

T = 0.02; % Simulation duration (s)

t = 0:1/fs:T;

%% Message Signal

m = Am * cos(2*pi*fm*t);

%% Frequency Deviation

delta_f = kf * Am;

%% Modulation Index

beta = delta_f / fm;

%% FM Signal

fm_signal = Ac * cos(2*pi*fc*t + beta * sin(2*pi*fm*t));

%% Plot Message and FM Signals

figure;

subplot(2,1,1)

plot(t,m,'LineWidth',1.5)

```

title('Message Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

subplot(2,1,2)
plot(t,fm_signal,'LineWidth',1.5)
title('Frequency Modulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);

figure
plot(f,FM_fft,'LineWidth',1.5)
title('FM Signal Spectrum')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
grid on
xlim([fc-10000 fc+10000])

%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('-----\n');
fprintf('Carrier Frequency    = %.0f Hz\n',fc);

```

```

fprintf('Message Frequency    = %.0f Hz\n',fm);

fprintf('Message Amplitude    = %.2f V\n',Am);

fprintf('Frequency Sensitivity = %.0f Hz/V\n',kf);


fprintf('\nCALCULATED VALUES\n');

fprintf('-----\n');

fprintf('Frequency Deviation ( $\Delta f$ ) = %.0f Hz\n',delta_f);

fprintf('Modulation Index ( $\beta$ )    = %.2f\n',beta);


%% Carson's Rule Bandwidth

BW = 2 * (delta_f + fm);

fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);

```

OUTPUT:

